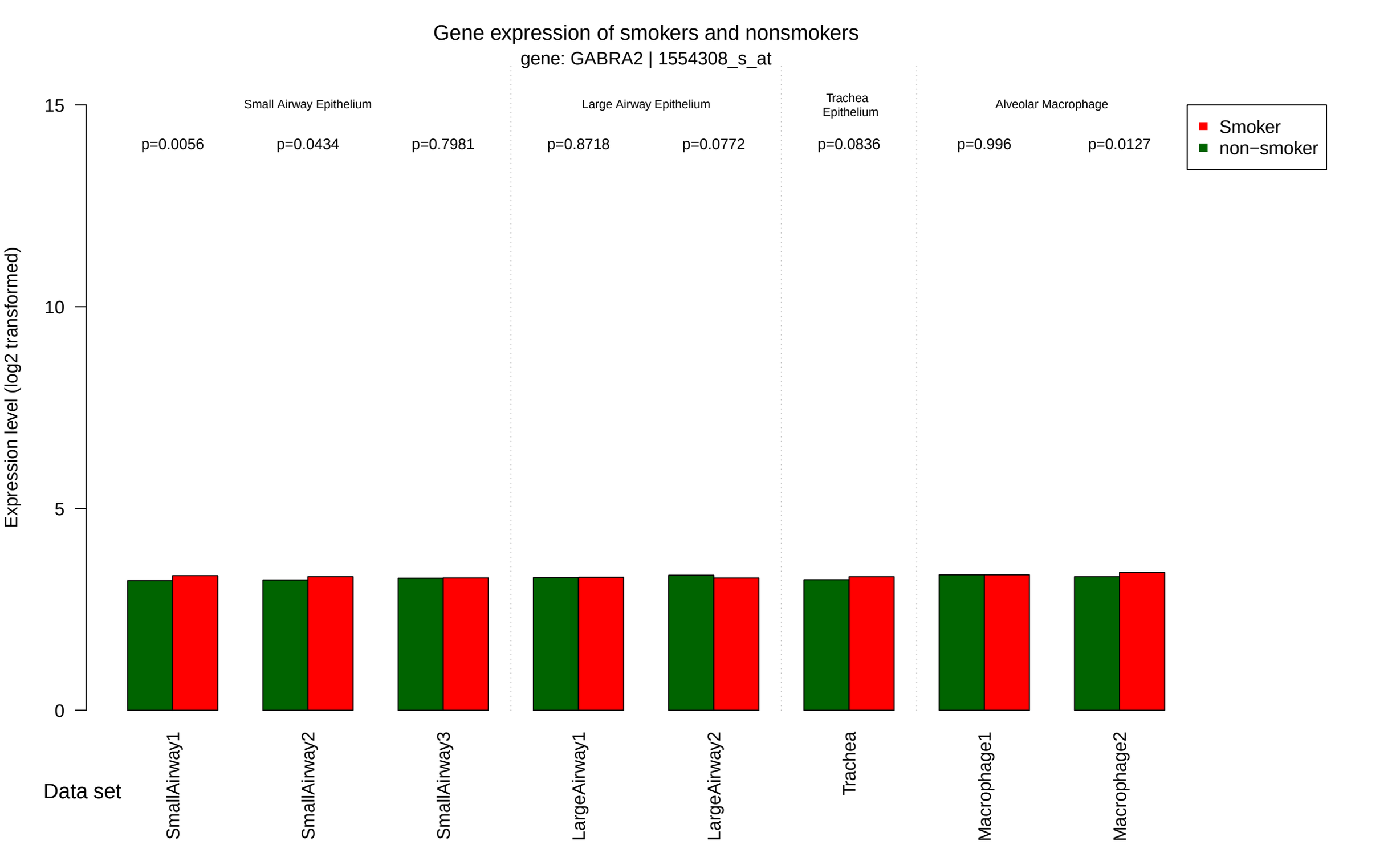
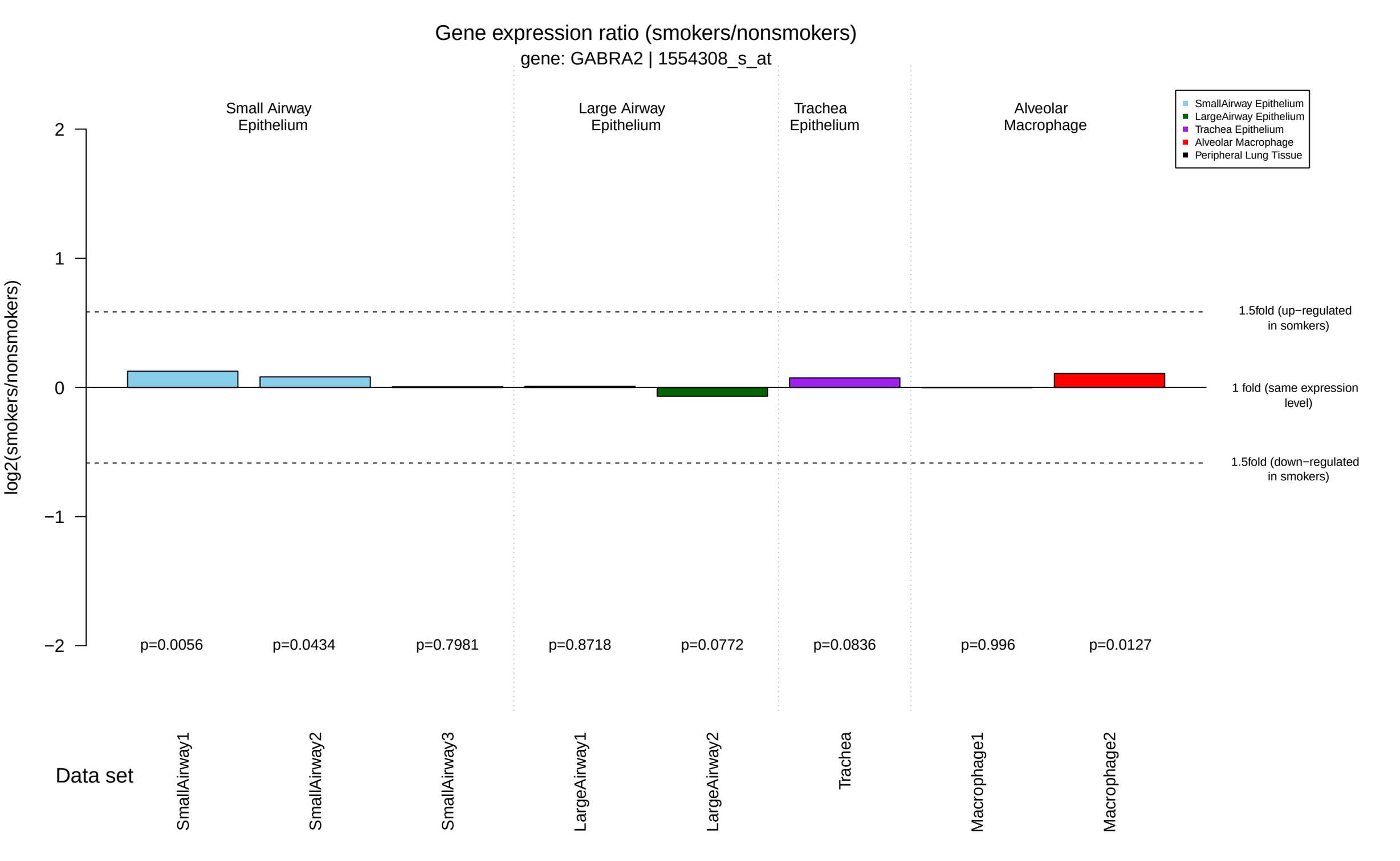
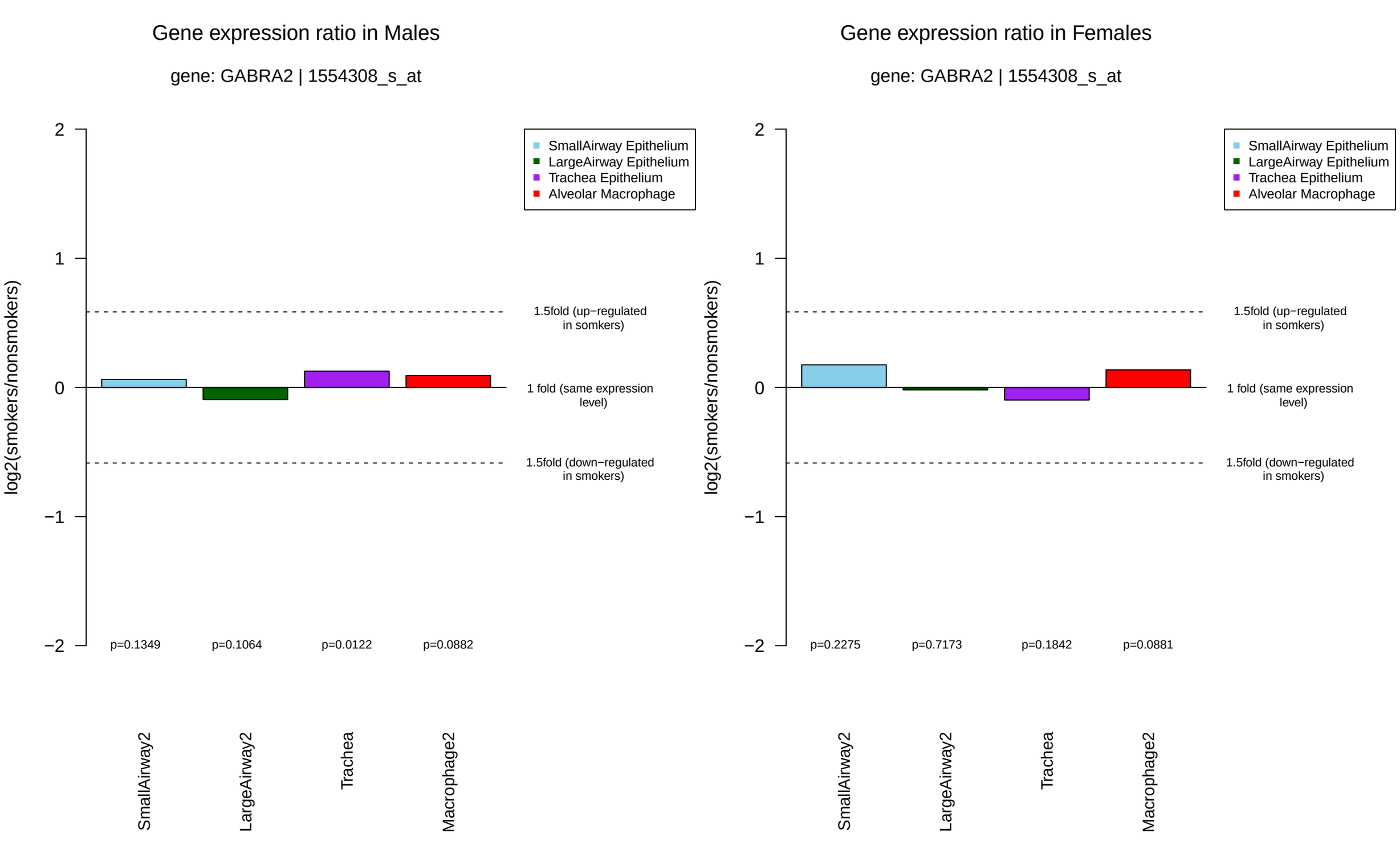
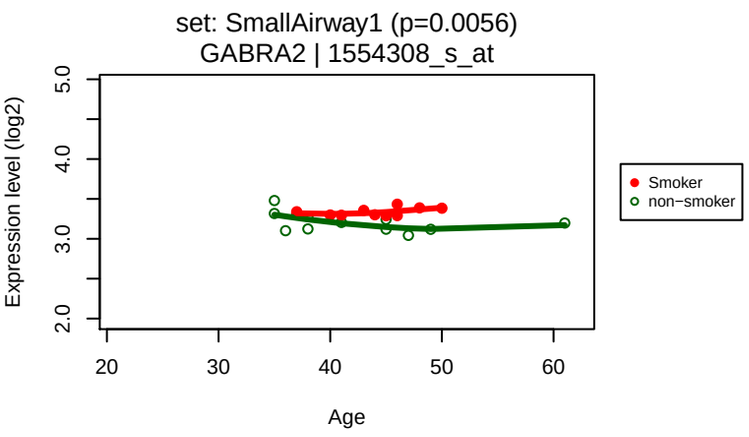
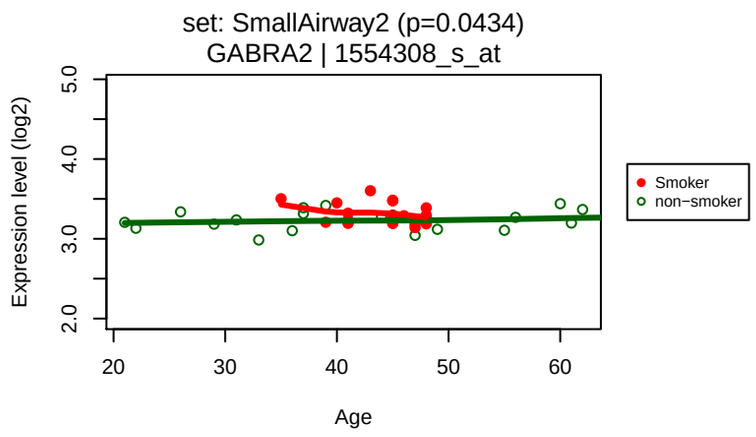
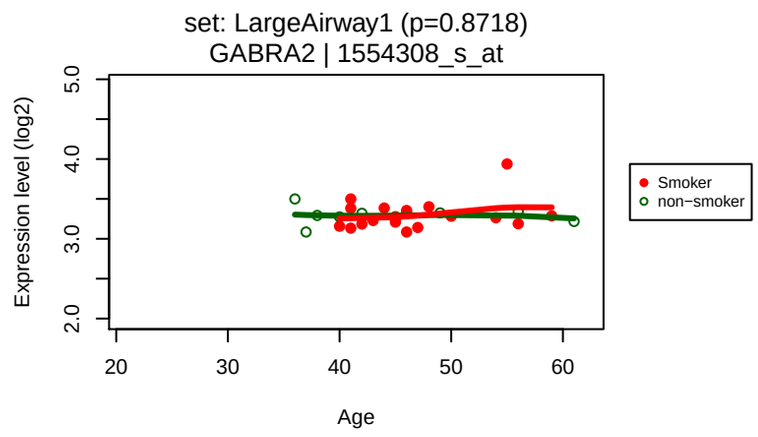

Gene expression of smokers and nonsmokers
gene: GABRA2 | 1554308_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.0056
p=0.0434
p=0.7981
p=0.8718
p=0.0772
p=0.0836
p=0.996
p=0.0127
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

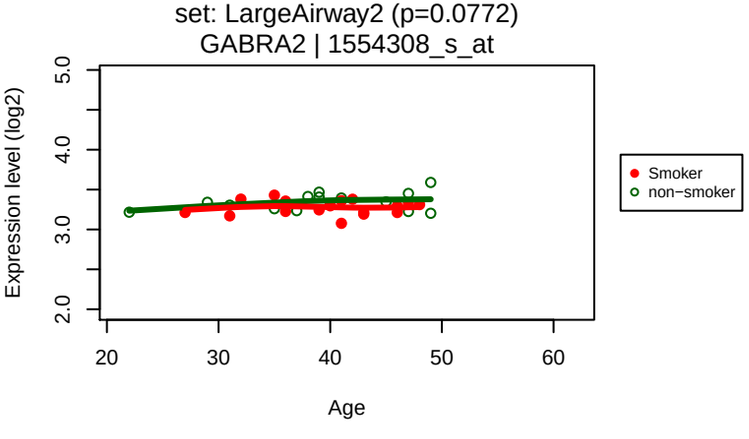

Gene expression ratio (smokers/nonsmokers)
gene: GABRA2 | 1554308_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0056
p=0.0434
p=0.7981
p=0.8718
p=0.0772
p=0.0836
p=0.996
p=0.0127
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

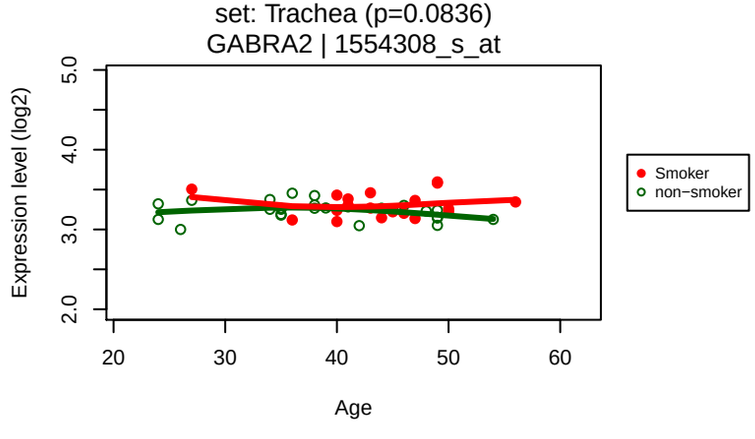

Gene expression ratio in Males
gene: GABRA2 | 1554308_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.1349
p=0.1064
p=0.0122
p=0.0882
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: GABRA2 | 1554308_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.2275
p=0.7173
p=0.1842
p=0.0881
SmallAirway2
LargeAirway2
Trachea
Macrophage2

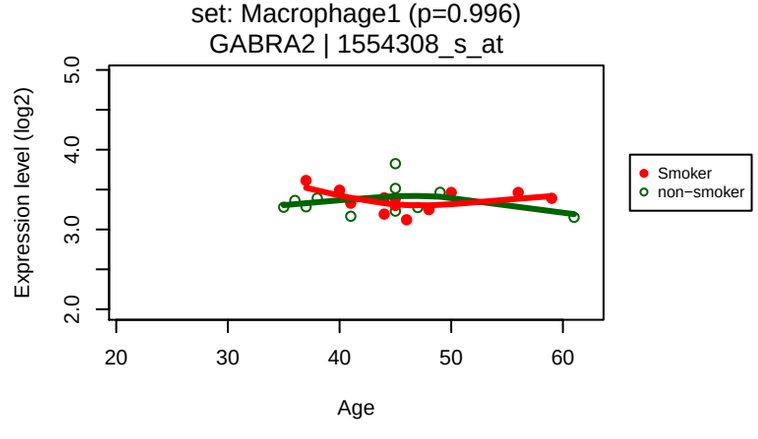

set: SmallAirway1 (p=0.0056)
GABRA2 | 1554308_s_at
Expression level (log2)
Age
Smoker
non−smoker

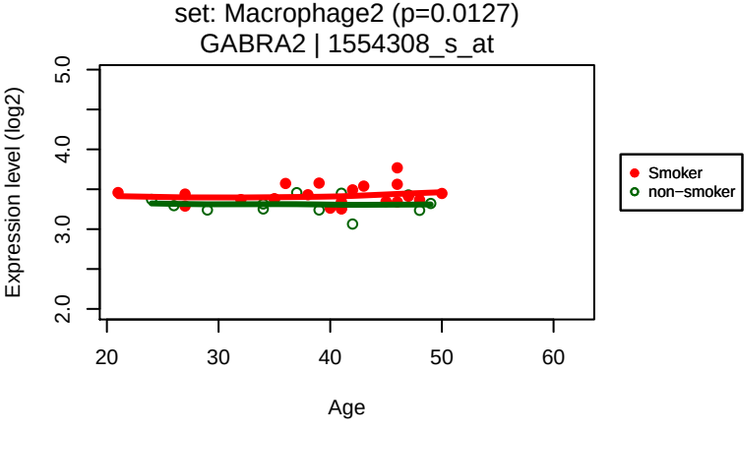

set: SmallAirway2 (p=0.0434)
GABRA2 | 1554308_s_at
Expression level (log2)
Age
Smoker
non−smoker

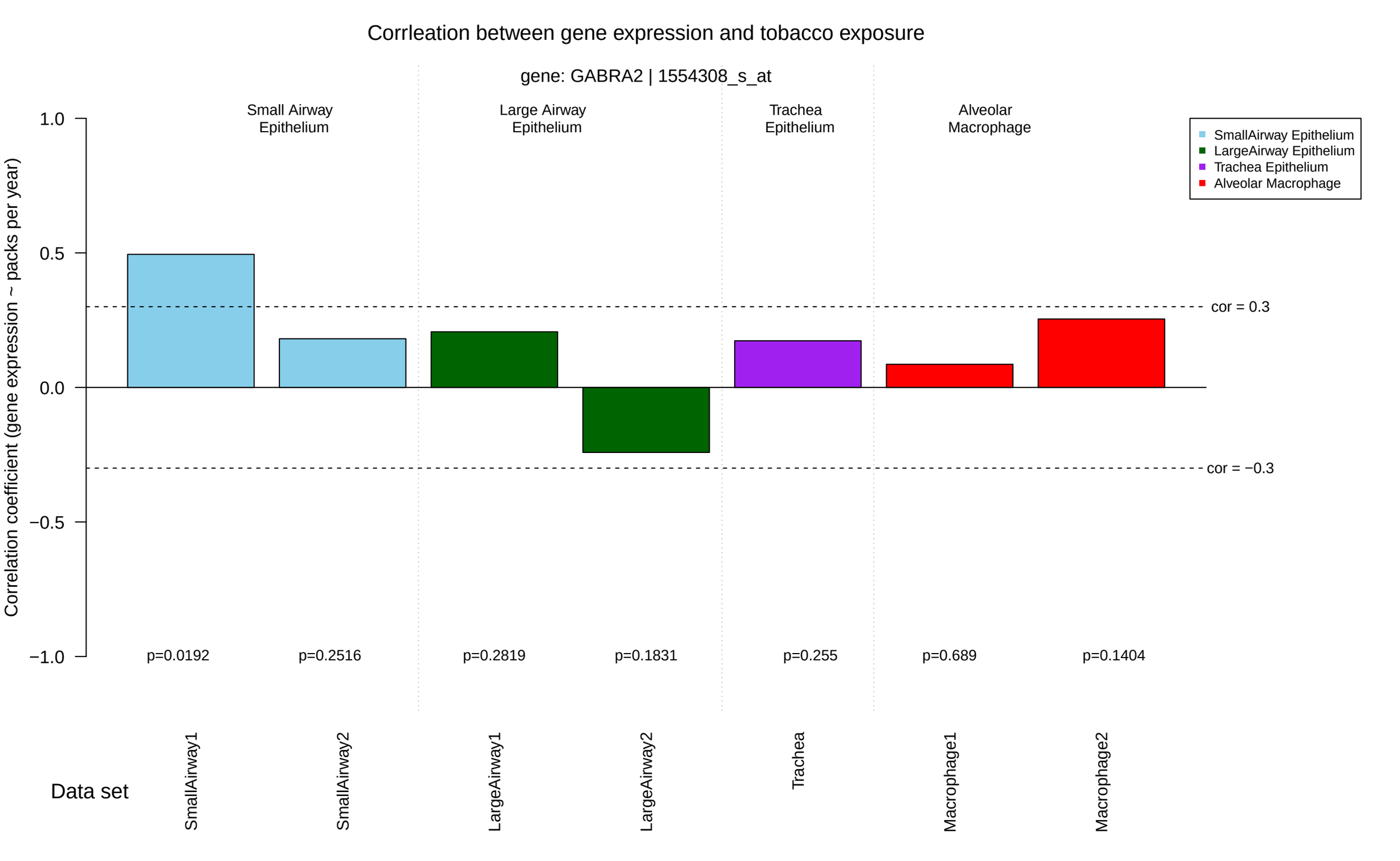

set: LargeAirway1 (p=0.8718)
GABRA2 | 1554308_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.0772)
GABRA2 | 1554308_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.0836)
GABRA2 | 1554308_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.996)
GABRA2 | 1554308_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.0127)
GABRA2 | 1554308_s_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: GABRA2 | 1554308_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.0192
p=0.2516
p=0.2819
p=0.1831
p=0.255
p=0.689
p=0.1404
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2